Hypothesis Testing

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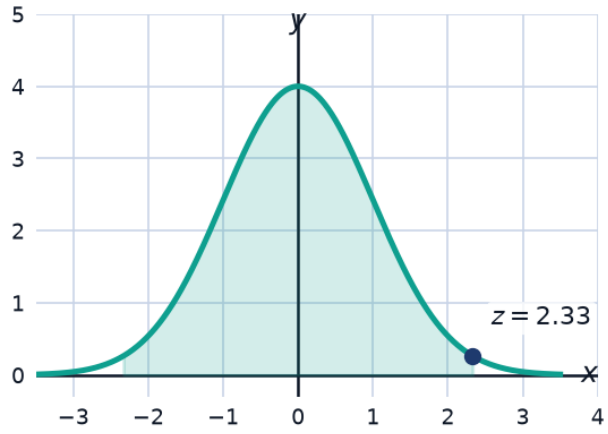
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: The logic of hypothesis testing

State the null and alternative hypotheses for testing whether a new fertilizer increases the mean crop yield μ, given a historical mean $\mu_0 = 50$ kg. Classify the test as one-tailed or two-tailed.

2 [Foundational] • Target: The logic of hypothesis testing

Explain the difference between a one-tailed and a two-tailed hypothesis test, giving an example scenario for each.



3 [Foundational] • Target: The logic of hypothesis testing

A test has $H_0 : \mu = 75$ vs $H_a : \mu \neq 75$, and the sample data leads to rejecting H_0 at the 5% significance level. State, without further calculation, whether this data would also lead to rejection at the 1% level, justifying your answer.

4 [Foundational] • Target: Hypothesis test for a population mean (known σ)

A machine fills bags with a claimed mean of 500g, with known population standard deviation $\sigma = 15$g. A sample of $n = 36$ bags has mean $\bar{x} = 494$g. Test at the 5% level whether the mean is less than 500g, using $z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$ and critical value -1.645.

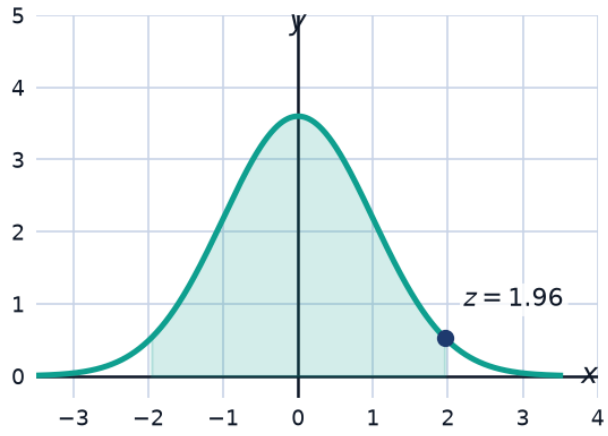
5 [Foundational] • Target: Hypothesis test for a population mean (known σ)

A population has known standard deviation $\sigma = 8$. A sample of $n = 64$ has mean $\bar{x} = 103$, against a claimed mean $\mu_0 = 100$. Test at the 1% level whether the mean exceeds 100, using critical value $z^* = 2.326$.

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6 [Foundational] • Target: Two-tailed tests for a population mean

A standardized test historically has mean $\mu_0 = 100$ with $\sigma = 15$. A sample of $n = 25$ students taking a new prep course has mean $\bar{x} = 107$. Test at the 5% level (two-tailed) whether the course changes the mean score, using critical values ± 1.96.



7 [Foundational] • Target: Two-tailed tests for a population mean

Using $z \approx 2.33$ from the previous problem, find the two-tailed p-value, given $\Phi(2.33) \approx 0.9901$, and confirm the conclusion.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Two-tailed tests for a population mean

A sample of $n = 50$ has mean $\bar{x} = 48.2$, against a claimed mean $\mu_0 = 50$ with $\sigma = 6$. Test at the 5% level (two-tailed) whether the mean differs from 50, using critical values ± 1.96.

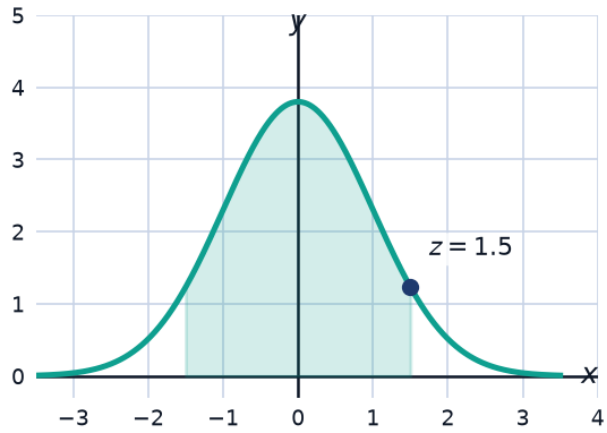
9 [Intermediate] • Target: Hypothesis test for a Poisson mean

A website historically crashes at a rate of $\lambda_0 = 4$ per month. After a software update, 1 crash is observed in a randomly selected month. Test at the 5% level whether the crash rate has decreased, using $P(X \leq 1) \approx 0.0916$ for $X \sim \text{Po}(4)$.

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10 [Intermediate] • Target: Hypothesis test for a Poisson mean

Explain why an exact Poisson probability, rather than a normal approximation, is used to test a hypothesis about a Poisson mean when λ_0 is small.



11 [Intermediate] • Target: Further Poisson mean tests

A call center historically receives calls at $\lambda_0 = 6$ per hour. During a promotional event, 12 calls are received in one hour. Test at the 5% level whether the call rate has increased, using $P(X \geq 12) \approx 0.0201$ for $X \sim \text{Po}(6)$.

12 [Intermediate] • Target: Further Poisson mean tests

A factory's defect rate is historically $\lambda_0 = 3$ per shift. During a trial of new equipment, 8 defects are observed in one shift. Test at the 5% level whether the defect rate has increased, using $P(X \geq 8) \approx 0.0119$ for $X \sim \text{Po}(3)$.

13 [Intermediate] • Target: Hypothesis test for a binomial proportion

A company claims 40% of its customers use its mobile app ($p_0 = 0.4$). In a sample of $n = 15$ customers, 10 report using the app. Test at the 5% level whether the true proportion is greater than 40%, using $P(X \geq 10) \approx 0.0338$ for $X \sim B(15, 0.4)$.

14 [Intermediate] • Target: Hypothesis test for a binomial proportion

Explain why the binomial hypothesis test uses the exact binomial distribution rather than a normal approximation when n is small.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

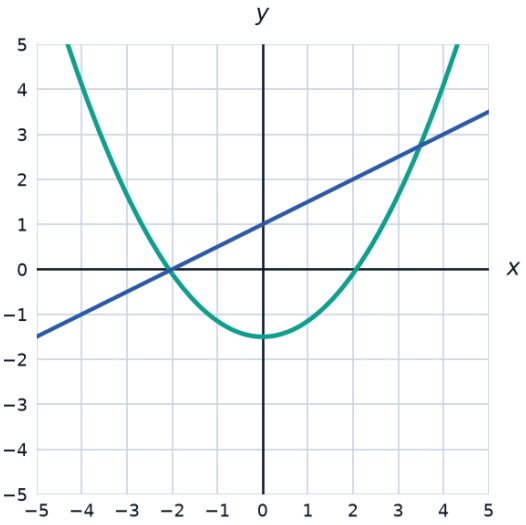
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15 [Challenging] • Target: Further binomial proportion tests • Cross-Lesson: 2.01 Exponentials & Logarithms

A coin is tested for fairness ($p_0 = 0.5$). In 20 tosses, 6 heads are observed. Test at the 5% level (two-tailed) whether the coin is biased, using $P(X \le 6) \approx 0.0577$ for $X \sim B(20, 0.5)$ (doubled for a two-tailed test).

16 [Challenging] • Target: Further binomial proportion tests • Cross-Lesson: 3.01 Advanced Trigonometry

A politician claims 60% of voters support a new policy ($p_0 = 0.6$). In a sample of 12 voters, only 3 support it. Test at the 5% level whether support is less than 60%, using $P(X \le 3) \approx 0.0153$ for $X \sim B(12, 0.6)$.



17 [Challenging] • Target: Type I and Type II errors, and significance level • Cross-Lesson: 4.01 Differentiation

In the website crash-rate test, describe what a Type I error would mean in context, and what a Type II error would mean.

18 [Challenging] • Target: Type I and Type II errors, and significance level • Cross-Lesson: 5.01 Integration

Explain what is meant by the significance level of a hypothesis test, and state the probability of a Type I error in a test conducted at the 5% significance level.

19 [Challenging] • Target: Type I and Type II errors, and significance level • Cross-Lesson: 6.01 Sequences & Series

A researcher conducts a test at $\alpha = 0.10$ instead of the more common $\alpha = 0.05$. Explain how this affects the probability of Type I and Type II errors, compared to using $\alpha = 0.05$.

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20 [Challenging] • Target: A well-designed word problem: testing a new manufacturing process • Cross-Lesson: 7.01 Vectors

This question has four parts. A factory's current process produces components with mean breaking strength $\mu_0 = 250$ N and known $\sigma = 20$ N. A new process gives a sample of $n = 40$ components with mean $\bar{x} = 257$ N. (a) State H_0 and H_a for testing whether the new process increases mean breaking strength (one-tailed). (b) Compute the test statistic z. (c) Using critical value $z^* = 1.645$ (5% level), state the conclusion. (d) Quality control also monitors defects; the historical rate is $\lambda_0 = 5$ per batch, but a batch from the new process has only 1 defect. Using $P(X \le 1) \approx 0.0404$ for $X \sim \text{Po}(5)$, test at the 5% level whether the new process has reduced the defect rate.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: The logic of hypothesis testing

For a hypothesis test at the 5% (two-tail) significance level, find the critical z-value.

a. 2.576

b. 1.645

c. 1.96

d. 1.282

22 [Foundational] • Target: Hypothesis test for a population mean (known σ)

A hypothesis test at the 1% significance level gives a p-value of 6%. What is the decision?

a. Test is inconclusive

b. Reject H_0

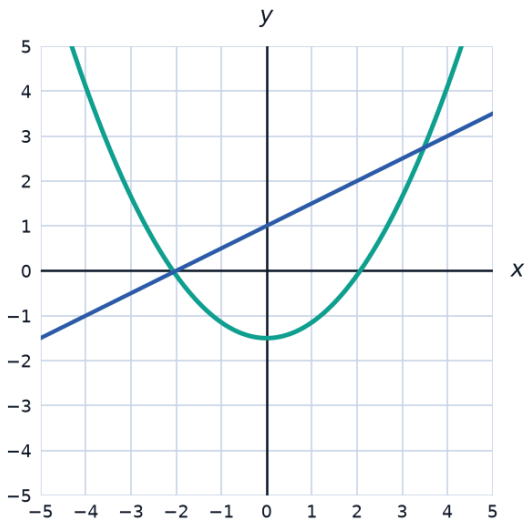
c. Increase the sample size

d. Do not reject H_0

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23 [Foundational] • Target: Two-tailed tests for a population mean

For a hypothesis test at the 5% (one-tail) significance level, find the critical z-value.



a. 1.96

b. 1.645

c. 1.282

d. 2.576

24 [Foundational] • Target: Hypothesis test for a Poisson mean

A test of $H_0 : \mu = 47$ uses a sample of size $n = 9$, with $\sigma = 8$ (known) and $\bar{x} = 55$. Find the test statistic $z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$.

a. 3

b. -3

c. 4

d. 9

25 [Foundational] • Target: Further Poisson mean tests

A two-tail hypothesis test has critical value $z = \pm 2.326$ and test statistic $z = 2$. What is the decision?

a. Accept H_1 unconditionally

b. Insufficient information

c. Reject H_0

d. Do not reject H_0

26 [Foundational] • Target: Hypothesis test for a binomial proportion

A one-tail (lower) hypothesis test has critical value $z = -2.326$ and test statistic $z = -1$. What is the decision?

a. Do not reject H_0

b. Reject H_0

c. Insufficient information

d. Accept H_1 unconditionally

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27 [Foundational] • Target: Further binomial proportion tests

A sample of size $n = 49$ has mean $\bar{x} = 44$, drawn from a population with $\mu_0 = 47$ and known $\sigma = 10$. Find the test statistic z, correct to 2 decimal places.

a. -2.10

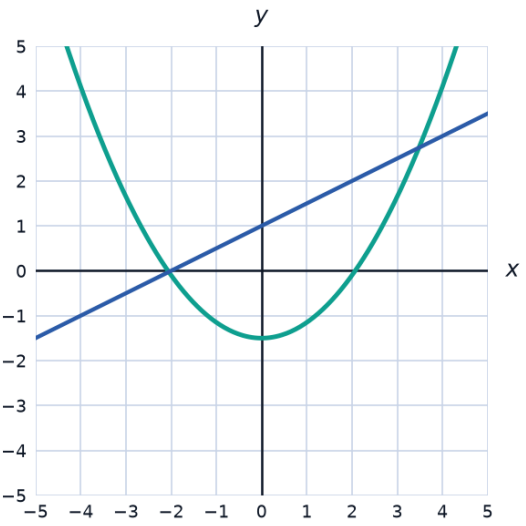
b. -3.60

c. -1.60

d. -1.10

28 [Foundational] • Target: Type I and Type II errors, and significance level

A population is modeled by $P(t) = P_0 \cdot 2^{t/10}$ with $P_0 = 200$. Find t when $P(t) = 400$.



a. 20

b. 10

c. 11

d. 1

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: A well-designed word problem: testing a new manufacturing process

Two lines have direction vectors $\mathbf{d_1} = (-2, 3, -5)$ and $\mathbf{d_2} = (4, -6, 10)$. What is the relationship between the two lines (assuming they are coplanar)?

a. neither parallel nor perpendicular

b. perpendicular

c. identical

d. parallel

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30 [Intermediate] • Target: The logic of hypothesis testing

A population is modeled by $P(t) = P_0 \cdot 2^{t/5}$ with $P_0 = 100$. Find $P(5)$.

a. 100

b. 102

c. 300

d. 200

31 [Intermediate] • Target: Hypothesis test for a population mean (known σ)

A population grows according to $\frac{dP}{dt} = kP$. If the population triples every 1 year, find k.

a. $\log(4)$

b. $1 + \log(3)$

c. $\log(3)$

d. $\log(3) + 2$

32 [Intermediate] • Target: Two-tailed tests for a population mean

A population grows according to $\frac{dP}{dt} = kP$. If the population triples every 2 years, find k.

a. $\frac{\log(4)}{2}$

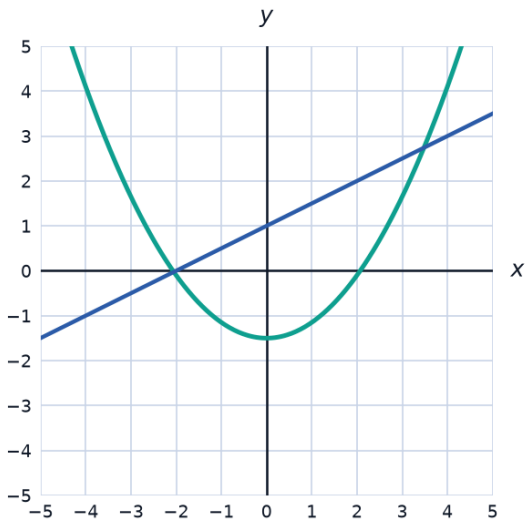
b. $2\log(3)$

c. $\frac{\log(3)}{2}$

d. $\frac{\log(3)}{2} + 1$

33 [Intermediate] • Target: Hypothesis test for a Poisson mean

Find the coefficient of x^2 in the binomial expansion of $(1 + x)^4$.



a. 6

b. 12

c. -6

d. $\frac{13}{2}$

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34 [Intermediate] • Target: Further Poisson mean tests

The scores on a standardized test are normally distributed with a mean of 100 and a standard deviation of 15. A new scoring algorithm is implemented, and the scores are transformed using the rule $Y = aX + b$, where X is the original score and Y is the new score. The new distribution of scores, Y, also follows a normal distribution with a mean of 120 and a standard deviation of 22.5. Calculate the value of a and the value of b.

a. a = 1.5, b = -30

b. a = -1.5, b = 270

c. a = 1.5, b = 30

d. a = 0.67, b = 53

35 [Intermediate] • Target: Hypothesis test for a binomial proportion

Line $l_1 : \mathbf{r} = (-3, 2, -2) + t(1, 3, 1)$ and line $l_2 : \mathbf{r} = (-7, 1, -5) + s(-3, 2, -2)$ intersect at one point. Find the y-coordinate of the point of intersection.

a. -1

b. -4

c. -3

d. 0

36 [Intermediate] • Target: Further binomial proportion tests

Find the coefficient of x^3 in the Maclaurin series of $\cos x$.

a. $-\frac{1}{6}$

b. $\frac{1}{6}$

c. $\frac{1}{3}$

d. 0

37 [Intermediate] • Target: Type I and Type II errors, and significance level

Line $l_1 : \mathbf{r} = (-7, -11, 3) + t(-1, -3, -1)$ and line $l_2 : \mathbf{r} = (-1, -5, 3) + s(-1, 1, 1)$ intersect at one point. Find the x-coordinate of the point of intersection.

a. 6

b. -3

c. -2

d. -4

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: A well-designed word problem: testing a new manufacturing process • Cross-Lesson: 5.01 Integration

Describe the locus $|z - (2i)| = 1$ in the Argand diagram. Find its radius.

a. 1

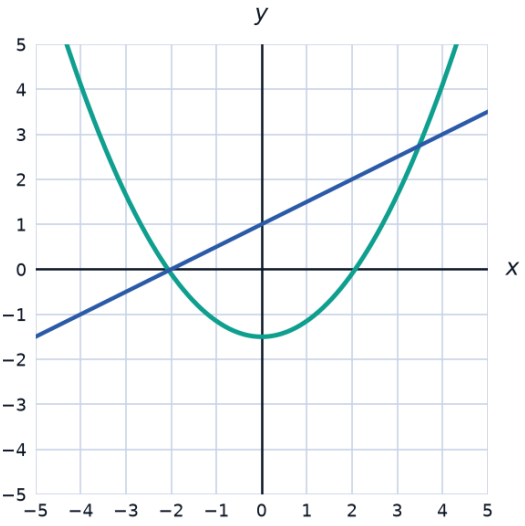
b. 4

c. 2

d. 5

39 [Challenging] • Target: The logic of hypothesis testing • Cross-Lesson: 6.01 Sequences & Series

A sequence satisfies $u_{n+1} = \frac{1}{2}u_n + 8$ with $u_1 = 5$. Find u_3.



a. $\frac{33}{4}$

b. $\frac{85}{4}$

c. 5

d. $\frac{53}{4}$

40 [Challenging] • Target: Hypothesis test for a population mean (known σ) • Cross-Lesson: 7.01 Vectors

Differentiate $y = (x)^2$ and evaluate $\frac{dy}{dx}$ at $x = 0$.

a. 4

b. 5

c. 1

d. 0

41 [Challenging] • Target: Two-tailed tests for a population mean • Cross-Lesson: 8.01 Algebra: Partial Fractions & Division

A geometric sequence has first term $a = 1$ and common ratio $r = 2$. Find the sum of the first 5 terms.

a. 31

b. -31

c. 32

d. 16

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42 [Challenging] • Target: Hypothesis test for a Poisson mean • Cross-Lesson: 9.01 Complex Numbers

Given that $(x + 3)$ is a factor of $P(x) = 4x^3 - 3x^2 - x + m$, find the value of m.

a. 132

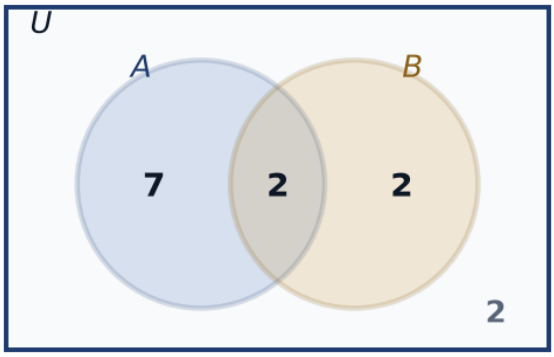
b. 129

c. 136

d. -132

43 [Challenging] • Target: Further Poisson mean tests • Cross-Lesson: 1.01 Functions: Modulus & Inverse

Line $l_1 : \mathbf{r} = (0, 7, -2) + t(-1, 1, -2)$ and line $l_2 : \mathbf{r} = (-1, 4, 6) + s(1, 1, -3)$ intersect at one point. Find the z-coordinate of the point of intersection.



a. 6

b. 4

c. 0

d. 1

44 [Challenging] • Target: Hypothesis test for a binomial proportion • Cross-Lesson: 10.01 Numerical Methods

Given $\tan\theta = \frac{5}{3}$ with θ in Quadrant III, state the sign of $\sec\theta$.

a. zero

b. negative

c. positive

d. undefined

45 [Challenging] • Target: Further binomial proportion tests • Cross-Lesson: 11.01 Differential Equations

Given $\tan\theta = -\frac{5}{3}$ with θ in Quadrant IV, state the sign of $\sec\theta$.

a. negative

b. zero

c. positive

d. undefined

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